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# LECTURE 3: Forms of Energy and Calculating Work
# Script: Maxwell-Boltzmann Demo
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# Description:
#   This notebook lets you interactively explore the
↪   Maxwell-Boltzmann speed distribution
#   and connect it to molecular kinetic energy.

# Learning goals
# - See how the distribution **shifts and broadens** as
↪   temperature increases
# - Compare how **molecular mass** changes the typical speeds
# - Verify numerically that
#  $\langle \frac{1}{2} m v^2 \rangle =$ 
↪  $\frac{3}{2} k_B T$ 

# Tip: Run cells top-to-bottom once. Then use the
↪ sliders/dropdowns to explore.
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import numpy as np
import math
import matplotlib.pyplot as plt

# Interactive widgets (works in JupyterLab/Notebook; in some
↪ environments you may need:
#   pip install ipywidgets
#   jupyter nbextension enable --py widgetsnbextension

import ipywidgets as widgets
from IPython.display import display, Markdown

# Physical constants
kB = 1.380649e-23 # Boltzmann constant, J/K
NA = 6.02214076e23 # Avogadro's number, 1/mol
R = 8.314462618 # Gas constant, J/(mol K)

# A small library of molecules (molar mass in g/mol)
species_db = {
    "Helium (He)": 4.0026,
    "Hydrogen (H2)": 2.01588,
    "Nitrogen (N2)": 28.0134,
    "Oxygen (O2)": 31.9988,
    "Carbon dioxide (CO2)": 44.0095,
    "Water vapor (H2O)": 18.01528,
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    "Argon (Ar)": 39.948,
}

# Functions to calculate
def molar_mass_to_molecule_mass(M_g_per_mol: float) -> float:
    """Convert molar mass (g/mol) to mass per molecule (kg)."""
    return (M_g_per_mol * 1e-3) / NA

def f_MB(v: np.ndarray, m: float, T: float) -> np.ndarray:
    """Maxwell-Boltzmann speed distribution f(v) for speeds v
    ↪ (m/s)."""
    pref = 4 * math.pi * (m / (2 * math.pi * kB * T))**(1.5)
    return pref * v**2 * np.exp(-(m * v**2) / (2 * kB * T))

def speeds_characteristic(m: float, T: float):
    """Most probable, mean, and RMS speeds."""
    v_mp = math.sqrt(2 * kB * T / m)
    v_mean = math.sqrt(8 * kB * T / (math.pi * m))
    v_rms = math.sqrt(3 * kB * T / m)
    return v_mp, v_mean, v_rms

display(Markdown("□ **Setup complete.** Use the interactive cell
↪ below."))

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□ **Setup complete.** Use the interactive cell below.

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# Interactive plot: effect of temperature and molecular mass
# * Temperature T shifts and broadens the distribution.
# * Molecular mass m shifts the distribution:
#   heavier molecules have lower typical speeds at the same T.
#
# The plot shows f(v) vs. v and marks:
# * v_mp: most probable speed (peak location)
# * <v>: mean speed
# * v_rms: root-mean-square speeds

# Checks at the end verify numerical calculations
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# Generate a plot for Nitrogen at 300 K first; then make it
↪ interactive
def plot_mb(T=300.0, species="Nitrogen (N2)", v_max=2000,
↪ npts=2000, show_fills=False):
    M = species_db[species]                # g/mol

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m = molar_mass_to_molecule_mass(M)      # kg per molecule

v = np.linspace(0, float(v_max), int(npts))
f = f_MB(v, m, float(T))

# characteristic speeds
v_mp, v_mean, v_rms = speeds_characteristic(m, float(T))

# numerical checks (integrals) – trapz is deprecated. If you
↪ get an error, update numpy.
area = np.trapezoid(f, v)  # should be ~1
v_mean_num = np.trapezoid(v * f, v)
v2_mean_num = np.trapezoid((v**2) * f, v)

KE_avg_num = 0.5 * m * v2_mean_num  # J per molecule
KE_avg_theory = 1.5 * kB * float(T)

# Plot
plt.figure(figsize=(8, 4.5))
plt.plot(v, f)
plt.axvline(v_mp, linestyle="--", linewidth=2)  # most
↪ probable
plt.axvline(v_mean, linestyle=":", linewidth=2) # mean
plt.axvline(v_rms, linestyle="-.", linewidth=2) # RMS

plt.xlabel("Speed, v (m/s)")
plt.ylabel("Probability density, f(v) (s/m)")
plt.title(f"Maxwell-Boltzmann Speed Distribution –
↪ {species}, T = {T:.0f} K")
plt.xlim(0, v_max)
plt.ylim(bottom=0)

if show_fills:
    mask = v <= v_mean
    plt.fill_between(v[mask], f[mask], alpha=0.2)

plt.legend(
    ["f(v)",
     "v_mp (most probable)",
     "<v> (mean)",
     "v_rms (RMS)"],
    loc="upper right",
    frameon=True
)

plt.show()

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# --- IMPORTANT: this must be inside the function so area,
↪ etc. exist ---
md = f"""### Numerical checks

- Normalization:
 $\int_0^{\infty} f(v) dv \approx \{area:.6f\}$ 

- Mean speed:
numerical  $\{v\_mean\_num:.2f\}$ ,  $\text{m/s}$ ,
theory  $\{v\_mean:.2f\}$ ,  $\text{m/s}$ 

- Mean squared speed:
numerical  $\langle v^2 \rangle \approx$ 
↪  $\{v2\_mean\_num:.2e\}$ ,  $\text{(m/s)}^2$ ,
theory  $\{3 \cdot k_B \cdot \text{float}(T) / m : .2e\}$ ,  $\text{(m/s)}^2$ 

### Average translational kinetic energy per molecule

- Numerical:
 $\langle \frac{1}{2} m v^2 \rangle \approx \{KE\_avg\_num:.3e\}$ ,  $\text{J}$ 

- Theory:
 $\frac{3}{2} k_B T = \{KE\_avg\_theory:.3e\}$ ,  $\text{J}$ 
"""

display(Markdown(md))

# Widgets + interactive output (unchanged)
T_slider = widgets.FloatSlider(value=300, min=50, max=2000,
↪ step=10,
                                description="T (K)",
                                continuous_update=False)
species_dd = widgets.Dropdown(options=list(species_db.keys()),
                                value="Nitrogen (N2)",
                                description="Species")
vmax_slider = widgets.IntSlider(value=2000, min=500, max=6000,
↪ step=100,
                                description="v_max",
                                continuous_update=False)
shade_toggle = widgets.Checkbox(value=False, description="Shade
↪  $v \leq <v>$ ")

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ui = widgets.VBox([
    widgets.HBox([T_slider, species_dd]),
    widgets.HBox([vmax_slider, shade_toggle])
])

out = widgets.interactive_output(
    plot_mb,
    {"T": T_slider, "species": species_dd, "v_max": vmax_slider,
    ↪  "show_fills": shade_toggle}
)

display(ui, out)
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VBox(children=(HBox(children=(FloatSlider(value=300.0, continuous_update=False,

Output()